

Nathaniel Murcin

Aerospace Engineering

Georgia Institute of Technology

B.S. Aerospace Engineering
Expected May 2027

M.S. Aerospace Engineering
Expected May 2028



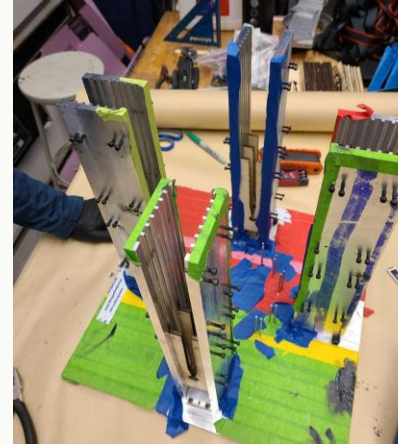
Selected engineering work

Blue Origin

Cryogenic systems and validated tooling

~60%

fewer tank-farm components

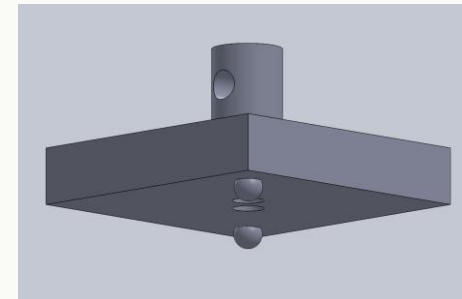


GTXR flight hardware

<0.1° error on all four fins

Pursuit Aerospace

Production tooling and qualified cleaning



Test engineering

Measured panel shear modulus to verify an ANSYS model

Cryogenic systems and manufacturing tooling

Ground-test fluid systems

Designed a cryogenic tank-farm P&ID, challenging requirements and simplifying heritage designs.

Sized piping, valves, and relief devices. Developed the BOM and coordinated procurement with suppliers.

~60%

reduction in component count

Adaptable gripping fixture

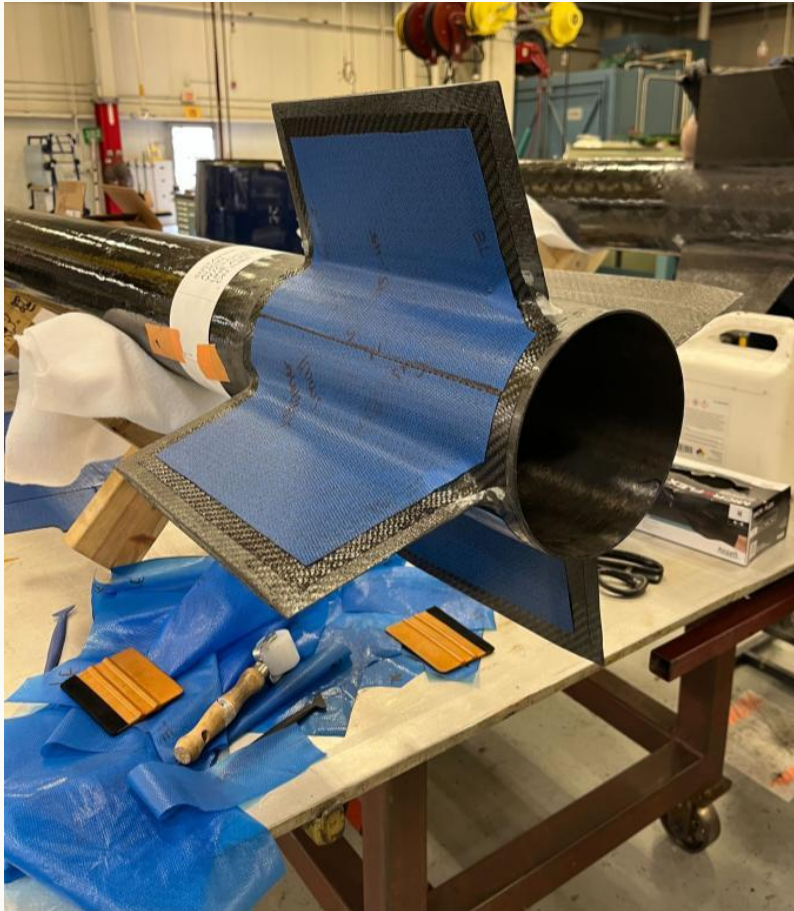
Designed six Creo versions to improve technician usability and prevent slip, part damage, and metal-chip FOD.

Validated with hand calculations and static FEA. Design passed PDR.

8 to 1

fasteners after fixture redesign

Fin selection within structural and machining limits



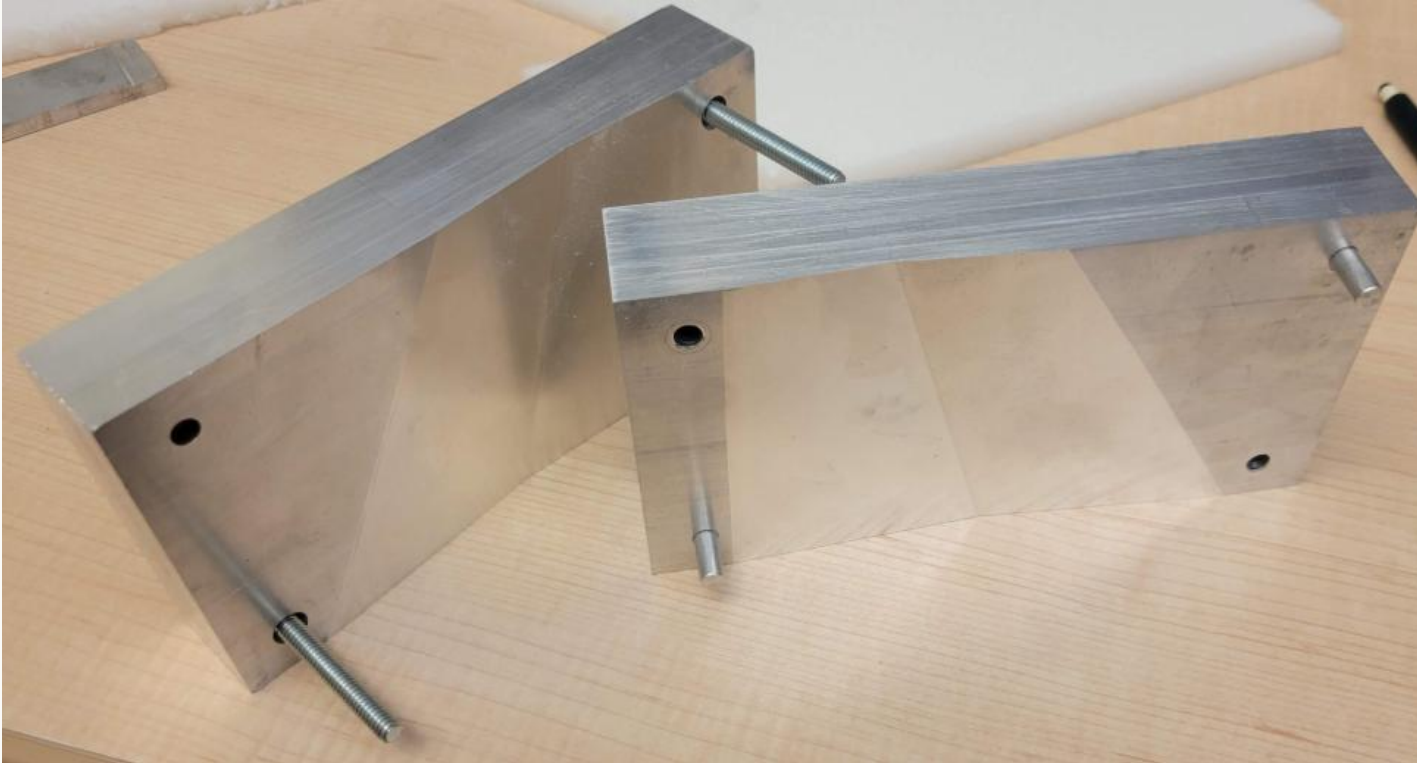
GTXR fin fabrication. Context photograph, not evidence of the selected material.

G10 sustainer fins

Built a decision matrix comparing materials, geometry, and manufacturing constraints.

The team selected G10 for the high-Mach sustainer. Booster choices were wood on one vehicle and G10 on another.

Thermoforming a candidate fin-core geometry



CNC-machined aluminum tooling for double-wedge PMI foam cores.

Co-developed and evaluated a thermoforming process for a future fin-core option.

0.28 mm

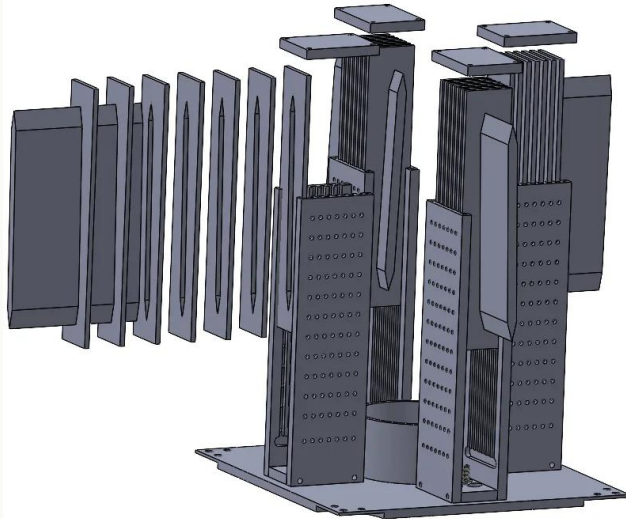
average global geometric error
measured with 3D metrology

Manufacturing feasibility research.
The PMI cores were not yet viable
for flight.

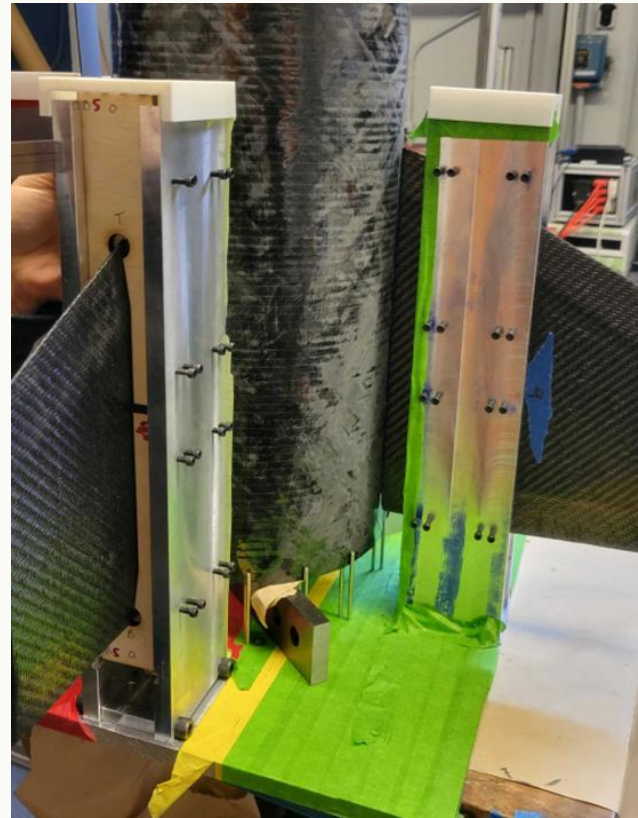
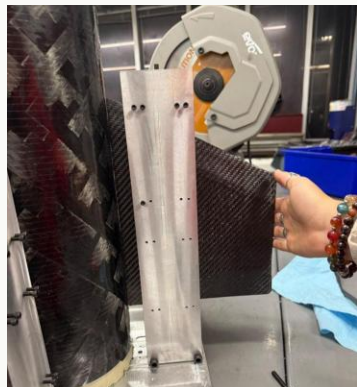
Co-author and presenter, AIAA 2026-112793

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Modular tooling for repeatable fin cant



CAD, machined components, and fixture assembly



Fixture on the airframe

Designed, CAM programmed, and machined a four-tower fixture for changing rocket diameters and fin profiles.

$<0.1^\circ$

error on all four fins, verified by independent metrology

Target cant: 1.6°
Laser-cut slots constrain each fin.

Production tooling and in-house cleaning

PHOTO NEEDED

Quick-change marking fixtures

Preferred: landscape
Show a fixture holding a representative approved part.

15 deployed fixtures

Designed and built dot-peen marking fixtures using GD&T and DFM.

6 s to 1 s

Average part load/unload time

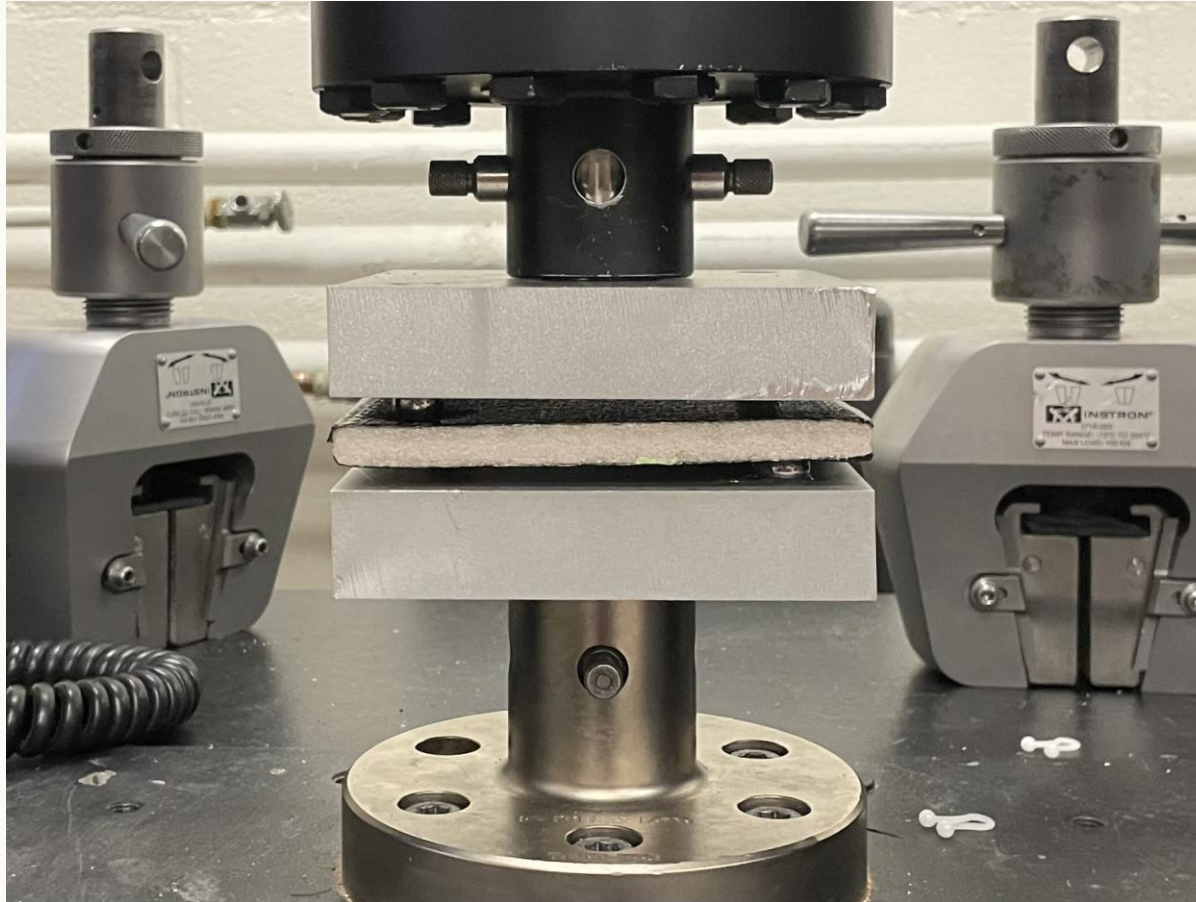
40 s to 2 s

Inspection time with Go/No-Go gauges

Requested: deployed tooling with the locating and clamping features visible.

Qualified in-house supercleaning removed 3 days and \$1,000/part in outsourced cleaning and shipping.

Plate-twist testing to verify the structural model



Physical test apparatus for in-plane shear modulus of sandwich panels.

Designed and executed the plate-twist test to characterize composite sandwich panels.

ANSYS model

Verified within a 5% acceptance criterion for future fin development.

Co-author and presenter
AIAA 2024-85690

Overnight recovery of an electrical passthrough

PHOTO NEEDED

Flight electrical passthrough

Preferred: landscape
Show the completed NPT body, wire exits, and sealing region.

A continuity failure threatened second-stage ignition. Designed and fabricated the replacement with the fin team.

14 hours

Design and fabrication during launch integration

2× MEOP

Hydrostatic test performed by the propulsion lead

The vehicle flew and the second stage ignited.

Mock launch rail from analysis to test



Full-height rail and vehicle at the mock test

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Guide hardware used in the rail assembly

Two-week build

Checked bending and buckling, completed CAD, and fabricated the rail in wood, steel, and aluminum.

Reacting-flow hardware and PIV processing

PHOTO NEEDED

Reacting jets-in-crossflow experiment

Preferred: landscape

Show approved multi-gas delivery hardware or the experimental setup.

Experimental support

Assembled regulated multi-gas delivery hardware for reacting jets-in-crossflow experiments.

Executed established MATLAB/PIV post-processing workflows.

Undergraduate researcher since January 2026

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